Boston, MA
davidw@tkins.me

David J. Watkins

GitHub: davidwatkins
LinkedIn: david-watkins
Website: davidjosephwatkins.com

Professional Experience

Research Lead — **June 2026 — Present**
Tutor Intelligence — *Watertown, MA*

- Foundation model research and training

Advisor — **May 2026 — Present**
Tola Capital — *Seattle, WA*

- Technical advisor on robotics and AI investments

Principal Research Lead — **September 2023 — June 2026**
Research Scientist — **Dec 2022 — August 2023**
RAI Institute (formerly Boston Dynamics AI Institute) — *Cambridge, MA*

- Pioneered handheld force-based data collection system for robot training, capturing force, vision, and proprioception for in-the-wild demonstrations
- Built video prediction system for robot policies using internet-scale video as prior (early 2023, before the wave); requires only limited real-world demonstrations
- Designed multimodal architectures combining novel sensors with internet-scale priors for improved behavioral execution (publication pending)
- Developed gradient-free RL enabling online learning with non-differentiable semantic functions (U.S. Patent pending, May 2025)
- Created task definition framework that makes human demonstrations more learnable for robots
- Built ROS 2 interfaces and controllers for custom grippers leveraging novel force/torque sensors
- Founded two research teams to scale innovations: Foundation Models (10+ researchers) and Capture (30 people)
- Built 320 H100 GPU infrastructure for large-scale training; created institute-wide policy execution framework
- Published at CoRL 2024, IJCAI 2024, ICML 2023; co-authoring “Elephants Don’t Write Sonnets” (MIT Press 2026)
- Established research partnerships with Google, Columbia, ETH Zurich, Agile Robots

Staff Software Engineer — **August 2022 — October 2022**
Righthand Robotics — *Somerville, MA*

- Worked on vision and planning team to improve pick and place performance for Rightpick systems
- Position ended due to company layoffs

CEO / Co-Founder, Odefi Inc. — **Mar 2019 — Jan 2022**
Columbia IBM Blockchain Accelerator — *New York, NY*

- Invented liquidity automation system for MakerDAO network; architected smart contract infrastructure
- Led technical architecture, product development, and investor pitches as part of Columbia IBM Blockchain Accelerator

Research Fellow — **Sep 2018 — July 2022**
Army Research Lab — *Aberdeen, MD*

- Won 1st place MineRL BASALT Competition (NeurIPS 2021): invented hierarchical agent combining learned and engineered components
- Invented mobile manipulation approaches without runtime localization (IROS 2022 Best Paper Finalist)
- Developed multimodal geometric learning fusing vision and tactile sensing for robotic grasping (ICRA 2019)
- Built simulation-to-real transfer pipelines for robotic grasping research

Engineering Intern — **Jun 2016 — Aug 2016**
Goldman Sachs — Margin Technology — *New York, NY*

- Built graph-based calculation prioritization system (Neo4j) optimizing ordering across 10K+ daily margin computations
- Developed Java/Angular.js system for dynamic prioritization weight adjustment

Engineering Intern — **May 2015 — Aug 2015**
Goldman Sachs — Valuations Technology — *New York, NY*

- Rebuilt FVA Gating Tool for derivatives valuations; implemented high-performance grid (100K+ rows, sub-second rendering)

Engineering Intern — **Jun 2014 — Sep 2014**
Streakfire LLC — *Wayland, MA and Dorado, PR*

- Produced ad campaign in Puerto Rico to promote technical accelerator for job creation
- Coordinated with local government to leverage publicity expertise and infrastructure

Technical Experience

Multiple View Performers for Shape Completion **Dec 2021 — Present**
Robotics at Google, Army Research Lab, Columbia University *New York, NY*

- Researched novel deep learning approach for multiple view completion without registering views
- Developed a process to leverage Performer attention layers developed by Google to encode multiple reconstruction images
- Work submitted to ICRA 2023

Data Driven Strand Simulation **Dec 2021 — Present**
Columbia University *New York, NY*

- Researched hair simulation algorithms to run in graph neural networks offering a 400% speedup
- Developed methods to compare python vs. C++ pytorch models
- In submission to TOG (ACM Transaction on Graphics)

MineRL Basalt Competition **Jul 2021 — Dec 2021**
Neurips 2021 *New York, NY*

- Researched the intersection of engineered and learned knowledge to develop an autonomous Minecraft agent using human demonstration data and won first place at MineRL Basalt at Neurips 2021 in collaboration with ARL and UMBC
- Researched learned visual navigation methods and developed a CNN state classifier using human-labeled data
- Work published at AAAI-Make 2022 and presented at Neurips 2021

Mobile Manipulation Leveraging Multiple Views **Jan 2020 — Oct 2022**
Columbia Robotics Lab *New York, NY*

- Researched deep learning approaches to mobile manipulation without localizing the robot at runtime
- Explored novel simulation based techniques for generating data with real-world scanned environments
- Improved previous navigation work via predicted panoramic target goals from nearby environment reconstruction
- Published to IROS 2022 and nominated for best paper in mobile manipulation

Learning from Electromyography Synergies to Grasp Novel Objects by Superquadric Representation **Jun 2018 — May 2020**
Columbia Robotics Lab *New York, NY*

- Collaborated with students to create a system for learning grasp synergies using a CTRL-Labs arm band and mapping the EMG signals to an anthropomorphic Seed robotic hand
- Helped develop an algorithm for planning and executing grasps using superquadric representations of novel objects with successful real-world demonstrations
- Work presented at Columbia Data Science Day 2019

Learning Your Way Without Map or Compass: Panoramic Target Driven Visual Navigation **Jan 2018 — Sep 2019**
Columbia Robotics Lab *New York, NY*

- Researched novel visual navigation methodology using RGBD panoramic target goals and behavioral cloning
- Developed a system architecture to embed images using an autoencoder and a policy model to control the robot
- Explored optimization strategies to develop training data from real-world environments without human intervention
- Work published to IROS 2020 and presented at NERC 2019

Multi-Modal Geometric Learning for Grasping and Manipulation **Sep 2017 — Sep 2019**
Columbia Robotics Lab *New York, NY*

- Incorporated tactile information to estimate shape geometry using vision and touch via deep learning
- Created a novel machine learned model for estimating shape via multi-modal information
- Work published to ICRA 2019 and to a RSS 2017 workshop

Human Robot Interface for Assistive Grasping **Jan 2017 — Dec 2018**
Columbia Robotics Lab and CUMC *New York, NY*

- Created a novel interface for enabling robot control for spinal cord injury patients using an sEMG device
- Benchmarked the sEMG interface against other modalities including an Amazon Echo and a toggle switch
- Work presented at Columbia Data Science Day 2018

Research in Bit Width Resolution **Jan 2016 — Dec 2016**
Columbia University *New York, NY*

- Worked with Professor Stephen Edwards at Columbia University to add Z3's SMT framework to an existing compiler project in order to resolve variable bit widths at compile time

Research in Shape Completion **Sep 2016 — May 2017**
Columbia Robotics Lab *New York, NY*

- Worked with Professor Peter Allen in the Columbia Robotics Lab at Columbia University to optimize an existing platform utilizing CUDA
- Evaluated the ability to utilize a semantic pre-processor to identify objects in a scene to be completed using the existing tool

Research in Data Visualization **Sep 2015 — Dec 2015**
Columbia University *New York, NY*

- Worked with Professor John Kender at Columbia University to provide visualization of the correlation between visual and textual memes in online video data and provided an analysis on the most effective ways of visualizing co-clustered data

Research in the Production of Litecoin ASICs **Jan 2014 — May 2014**
Columbia University *New York, NY*

- Cooperated with Professor Simha Sethumadhavan on the feasibility of producing Litecoin Mining ASICs
- Independently designed all of the ASIC schematics, performed cost-benefit analysis of the ASIC and maintained knowledge on which crypto-currencies were most profitable at any time

Written Values Affirmation Intervention to Identify the Unique Linguistic Features of Stigmatized Groups **Sep 2014 — Jan 2015**
LIRSM *New York, NY*

- Responsible for developing web application in Node.js and Mongo to easily add and retrieve information from csv files
- Participated in IT work and assisted with sessions in informing individuals on using UNIX
- Developed strategies to acquire study data more efficiently and help audit costs on services

Walking in Their Shoes: Poverty in America **Sep 2014 — Jan 2015**
Yale *New Haven, CT*

- Developed interface and game for flexible assignment of agency to test subjects to explore the impact that games have on depictions of poverty
- Worked with MongoDB and Angular.js to create flexible tiled gameplay

Publications

1. Surendran, Vidullan, Neehar Peri, and **David Watkins**. "Bridging Handheld and Teleoperated Supervision for Contact-Rich Manipulation via State-Gated Experts." Conference on Robot Learning (CoRL), 2026 (pending acceptance). https://nperi-rai.github.io/bridge-project/

2. Kapoor, Kunal, Shiraz Khan, Joseph McCalmon, Jesse Michel, Arif Mohammed, Katerina Nikiforova, Adam Oppenheimer, Victor Szabo, and **David Watkins**. "Maintaining Demonstration Quality in a 100-Robot Teleoperation Pipeline." Workshop on “It’s the Demos: A Deep Look at the Role of Demonstration Quality in Imitation-Based Robot Manipulation,” Robotics: Science and Systems (RSS), Sydney, Australia, 2026.

3. Tellex, Stefanie, and **David Watkins**. "Elephants Don’t Write Sonnets: The Grounded Turing Test for Embodied AI." 2025. https://h2r.cs.brown.edu/wp-content/uploads/tellexwatkins2026.pdf

4. Shang, Jinghuan, Karl Schmeckpeper, Brandon B. May, Maria Vittoria Minniti, Tarik Kelestemur, **David Watkins**, and Laura Herlant. ‘Theia: Distilling Diverse Vision Foundation Models for Robot Learning’. In 8th Annual Conference on Robot Learning, 2024. https://openreview.net/forum?id=ylZHvlwUcI.

5. Cohen, Vanya, Jason Xinyu Liu, Raymond Mooney, Stefanie Tellex, and **David Watkins**. "A Survey of Robotic Language Grounding: Tradeoffs Between Symbols and Embeddings." arXiv preprint arXiv:2405.13245 (2024).

6. Novoseller, Ellen, Vinicius G. Goecks, **David Watkins**, Josh Miller, and Nicholas Waytowich. "DIP-RL: Demonstration-Inferred Preference Learning in Minecraft." arXiv preprint arXiv:2307.12158 (2023).

7. Milani, S., Kanervisto, A., Ramanauskas, K., Schulhoff, S., Houghton, B., Mohanty, S., ... & Shah, R. (2023). Towards Solving Fuzzy Tasks with Human Feedback: A Retrospective of the MineRL BASALT 2022 Competition. arXiv preprint arXiv:2303.13512.

8. Choromanski, K.M., Sehanobish, A., Lin, H., Zhao, Y., Berger, E., Parshakova, T., Pan, A., **Watkins, D.**, Zhang, T., Likhosherstov, V. and Chowdhury, S.B.R., 2023, July. Efficient graph field integrators meet point clouds. In International Conference on Machine Learning (pp. 5978-6004). PMLR.

9. Hu, Jiaheng, **David Watkins**, and Peter Allen. "Teleoperated Robot Grasping in Virtual Reality Spaces." arXiv preprint arXiv:2301.13064 (2023).

10. **Watkins-Valls, D.**, Allen, P., Choromanski, K., Varley, J., & Waytowich, N. (2022). Multiple View Performers for Shape Completion. arXiv preprint arXiv:2209.06291.

11. **Watkins, David Joseph.** (2022). Learning Mobile Manipulation. Columbia University. https://doi.org/10.7916/V9YM-TQ84

12. **Watkins-Valls, D.**, Maia H., Varley J., Seshadri M., Sanabria J., Waytowich, N., & Allen, P. (2022). Mobile Manipulation Leveraging Multiple Views. 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2022

13. Goecks, Vinicius G., et al. “Combining Learning from Human Feedback and Knowledge Engineering to Solve Hierarchical Tasks in Minecraft.” ArXiv:2112.03482 [Cs], Dec. 2021. arXiv.org, http://arxiv.org/abs/2112.03482. Accepted to AAAI-Make 2022.

14. **Watkins-Valls, D.**, Xu, J., Waytowich, N., & Allen, P. (2020). Learning your way without map or compass: Panoramic target driven visual navigation. 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2020
15. Akinola, Iretiayo, Zizhao Wang, Junyao Shi, Xiaomin He, Pawan Lapborisuth, Jingxi Xu, **David Watkins-Valls**, Paul Sajda, and Peter Allen. "Accelerated Robot Learning via Human Brain Signals." In 2020 IEEE International Conference on Robotics and Automation (ICRA), pp. 3799-3805. IEEE, 2020.
16. Wu, B., Akinola, I., Gupta, A., Xu, F., Varley, J., **Watkins-Valls, D.**, & Allen, P. K. (2020). Generative Attention Learning: a “GenerAL” framework for high-performance multi-fingered grasping in clutter. Autonomous Robots, 1-20.
17. **Watkins-Valls, D.**, Varley, J. & Allen, P. Multi-Modal Geometric Learning for Grasping and Manipulation. 2019 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2019.
18. Abhi Gupta, Jingya Bi, Ashwin Jayaraman, Max Xu, **David Watkins** and Professor Peter Allen. “Learning from Electromyography Synergies to Grasp Novel Objects by Super Quadric Representation (Poster)” In: Columbia Data Science Day (2019).
19. Jacob Varley, **David Watkins-Valls**, and Peter Allen. “Multi-Modal Geometric Learning for Grasping and Manipulation (Poster)”. In: Columbia Data Science Day (2018).
20. Jacob Varley, **David Watkins**, and Peter Allen. “Visual-Tactile Geometric Reasoning (Abstract and Poster)”. In: Data-Driven Manipulation workshop, Robotics: Science and Systems (2017).
21. **David Watkins-Valls**, Chaiwen Chou, Caroline Weinberg, Jacob Varley, Lynne Weber, Adam Blanchard, Peter Allen, Joel Stein "Human Robot Interface for Assistive Grasping (Poster)". In: New England Manipulation Symposium (2017).
22. **David Watkins-Valls** "Scrypt Mining With ASICs" (2014).

Writing

1. **Watkins, David**, and Stefanie Tellex. "Diffusion Policies are Fancy Lookup Tables." *What to Tell the Robot*, July 21, 2026. https://whattotelltherobot.com/p/diffusion-policies-are-fancy-lookup
2. Tellex, Stefanie, and **David Watkins**. "We Got a New Nine." *What to Tell the Robot*, July 11, 2026. https://whattotelltherobot.com/p/we-got-a-new-nine
3. Tellex, Stefanie, and **David Watkins**. "I Need Space." *What to Tell the Robot*, July 6, 2026. https://whattotelltherobot.com/p/i-need-space
4. Tellex, Stefanie, and **David Watkins**. "Why Robotics is a Pre-Paradigm Field." *What to Tell the Robot*, June 6, 2026. https://whattotelltherobot.com/p/why-robotics-is-a-pre-paradigm-field
5. Tellex, Stefanie, and **David Watkins**. "I Got Sycophanted." *What to Tell the Robot*, May 20, 2026. https://whattotelltherobot.com/p/i-got-sycophanted
6. Tellex, Stefanie, and **David Watkins**. "How to Build Safe AI (Without Making the AI Safe)." *What to Tell the Robot*, May 13, 2026. https://whattotelltherobot.com/p/how-to-build-safe-ai-without-making
7. **Watkins, David**. "Getting a Grip on Robotic Data Collection." *RAI Institute Blog*, May 12, 2026. https://rai-inst.com/resources/blog/handheld-robotic-data-collection/
8. Tellex, Stefanie, and **David Watkins**. "The First Hit is Free." *What to Tell the Robot*, May 7, 2026. https://whattotelltherobot.com/p/the-first-hit-is-free
9. **Watkins, David**, and Stefanie Tellex. "What Babies Know That Robots Don’t." *What to Tell the Robot*, April 15, 2026. https://whattotelltherobot.com/p/what-babies-know-that-robots-dont
10. **Watkins, David**, and Stefanie Tellex. "Consciousoids." *What to Tell the Robot*, March 30, 2026. https://whattotelltherobot.com/p/consciousoids
11. Tellex, Stefanie, and **David Watkins**. "Elephants Still Don’t Play Chess." *What to Tell the Robot*, March 2, 2026. https://whattotelltherobot.com/p/elephants-still-dont-play-chess
12. Tellex, Stefanie, and **David Watkins**. "Radiate Love: a 2025 Retrospective." *What to Tell the Robot*, February 3, 2026. https://whattotelltherobot.com/p/radiate-love-a-2025-retrospective
13. **Watkins, David**, and Stefanie Tellex. "Satisfying a 31-Year-Old Karaoke Dream." *What to Tell the Robot*, January 16, 2026. https://whattotelltherobot.com/p/satisfying-a-31-year-old-karaoke

14. Tellex, Stefanie, and **David Watkins**. "Where the Curves Cross." *What to Tell the Robot*, January 9, 2026. https://whattotelltherobot.com/p/where-the-curves-cross
15. **Watkins, David**. "2025 in Review: The Year of Showing Up." *What to Tell the Robot*, December 31, 2025. https://whattotelltherobot.com/p/2025-in-review-the-year-of-showing
16. **Watkins, David**, and Stefanie Tellex. "The Deer Island Marvel." *What to Tell the Robot*, December 5, 2025. https://whattotelltherobot.com/p/the-deer-island-marvel
17. Tellex, Stefanie, and **David Watkins**. "Qualitative Simulation of Swine Production." *What to Tell the Robot*, November 19, 2025. https://whattotelltherobot.com/p/qualitative-simulation-of-swine-production
18. Tellex, Stefanie, and **David Watkins**. "The Grounded Turing Test." *What to Tell the Robot*, September 2, 2025. https://whattotelltherobot.com/p/the-grounded-turing-test
19. **Watkins, David**, and Stefanie Tellex. "From Physics to Embodied Intelligence." *What to Tell the Robot*, August 27, 2025. https://whattotelltherobot.com/p/from-physics-to-embodied-intelligence
20. Tellex, Stefanie, and **David Watkins**. "Taking a Quadruped Robot to School." *What to Tell the Robot*, August 20, 2025. https://whattotelltherobot.com/p/taking-a-quadruped-robot-to-school
21. Tellex, Stefanie, and **David Watkins**. "What to Tell the Robot." *What to Tell the Robot*, July 18, 2025. https://whattotelltherobot.com/p/what-to-tell-the-robot

Education

PhD in Computer Science, *Columbia University* — Sep 2017 — May 2022
Advisor: *Prof. Peter Allen*, Thesis: *Learning Mobile Manipulation*
Army Research Lab Research Fellow — Sep 2018 — July 2022
MPhil in Computer Science, *Columbia University* — Sep 2017 — May 2019
MS in Computer Science, *Columbia University*, 4.0 GPA — Sep 2016 — May 2017
CA Fellowship — Sep 2016 — Jan 2017
BS in Computer Science, *Columbia University*, 3.7 GPA — Sep 2012 — May 2016
Marian High School — Oct 2010 — May 2012
Class President
Salutatorian
National Honors Society

Awards

Best Paper in Mobile Manipulation Finalist, *Kyoto, Japan* — Oct 2022
Selected for my work titled Mobile Manipulation Leveraging Multiple Views submitted to IROS 2022
International House Member, *New York, NY* — Sep 2016 — May 2018
Competitively selected scholars and young professionals from around the world who are challenged to become globally-minded leaders
CA Fellowship, *Columbia University* — Sep 2016 — Dec 2016
MS students who have proven themselves to be exceptional will receive paid tuition and stipend
Goldman Sachs Code Golf Champion, *New York, NY* — Aug 2016
Awarded for shortest possible source code that implements a certain algorithm
Dean's List, *Columbia University* — Spring 2013, Spring 2015, Spring 2016
A list of students recognized for academic achievement during a semester by the dean of the college they attend
Residential Incubator Fellow, *Columbia University* — Sep 2012 — May 2014
Students interested in entrepreneurship participating in a student incubator

Activities

Tactile Sensing for Robotic Foundation Models Workshop Co-Organizer, RSS — 2026
New England Manipulation Symposium (NEMS) Co-Organizer — 2025
IJRR Reviewer — 2023
NSF Reviewer — 2023
ICRA Paper Reviewer — 2022, 2025
Computer Science Student Faculty Representative — 2018 — 2020
IEEE RA-L Paper Reviewer — 2021, 2024, 2025
IROS Paper Reviewer — 2018, 2021, 2023, 2024
CoRL Paper Reviewer — 2020, 2024

Skills

Languages	Python, C++, CUDA, ROS 2, PyTorch, Bash, LaTeX, SQL
ML/AI	Foundation Models, Transformers, Diffusion Models, Policy Learning, Distributed Training
Infrastructure	Kubernetes, Docker, H100/A100 Clusters, Slurm, Ray, Git, CI/CD
Robotics	Manipulation, Navigation, SLAM, Simulation (PyBullet, MuJoCo, Isaac), ROS 2
Leadership	Team Building (0→13+ FTEs), Hiring (200+ interviews), Strategic Partnerships
Communication	English, Spanish

Volunteer Work

Multisensory Reading Centers of Puerto Rico, *San Juan, PR* Sep 2017 — Present
Over 150 hours of volunteer service by performing IT help to provide access to effective literacy instruction for struggling readers

Watkins-Valls Family Foundation, *Boston, MA* Sep 2013 — Present
Providing scholarships and academic support to underprivileged students in Massachusetts, New York, and Puerto Rico

Pine St. Inn, *Boston, MA* Dec 2010 — May 2012
Over 25 hours of volunteer service at a soup kitchen for the homeless

Sisters of St. Joseph, *Cambridge, MA* Dec 2010 — May 2012
Over 25 hours of volunteer service through entertaining and assisting retired nuns

Presentations

What Data Robots Need: From Co-Designed Handheld Capture to Fleet-Scale Flywheels, *Extropic, Waltham, MA* Jul 2026
Invited talk on robot data collection, from co-designed handheld force-based capture to fleet-scale data flywheels, presented to the Extropic team

Tactile Sensing for Robotic Foundation Models Workshop Co-Organizer, *RSS 2026, Sydney* Jul 2026
Co-organized workshop on the role of tactile sensing in robotic foundation models and large-scale data collection

What Data Robots Need: From Co-Designed Handheld Capture to Fleet-Scale Flywheels, *RSS 2026 Workshop (It's the Demos), Sydney* Jul 2026
Invited workshop talk tracing robot data collection from co-designed handheld force-based capture to fleet-scale data flywheels, drawing on work at the RAI Institute and Data Factory 1 (DF1)

Why Robots? Data-Driven Manipulation for the World's Hardest Sorting Problem, *Brown University* Feb 2026
Guest lecture in Stefanie Tellex's class on robotic manipulation for landfill restoration, covering multimodal sensing, force-based data collection, and the capability gap between warehouse and unstructured-environment robots

New England Manipulation Symposium (NEMS) 2025 Co-Organizer, *MIT* Apr 2025
Co-organized conference with Lael Odhner and Kaitlyn Becker; coordinated paper acceptances, speaker scheduling, and venue logistics

The Future of Intelligent Robotics, *Dr. Waku YouTube Interview* 2025
Featured interview discussing robotics' "ChatGPT moment" and the future of embodied AI

Lions in AI Panel, *Columbia Alumni Association of Boston* 2025
Panel discussion on AI and robotics for Columbia alumni community

GTC 2025 Summary, *RAI Institute* Mar 2025
Presented key takeaways from NVIDIA GPU Technology Conference to institute leadership

Theia: Distilling Diverse Vision Foundation Models, *CoRL 2024, Munich* Nov 2024
Published work on foundation model distillation for robot learning at Conference on Robot Learning

Foundation Models for Robotics, *Duke University* Feb 2024
Presentation on state of the art foundation models for robotic learning

Helios: High Dimensional Predictive Foundation Models as Knowledge Transfer Systems, *Brown University* Jul 2023
Building large scale video predictive foundation models for robotics at the AI Institute

From One to Many: How to leverage multiple views in shape completion, *Robotics at Google* Dec 2022
A discussion on how to leverage multiple views for shape completion presented to researchers at Robotics at Google

Learning Mobile Manipulation, *Harvard - Harvard Biorobotics Laboratory* Nov 2022
Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Robert Howe's lab

Learning Mobile Manipulation, *Tufts - Human Robotics Interaction Laboratory* Nov 2022
Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Matthias Scheutz's lab

Learning Mobile Manipulation, *MIT - Improbable AI Lab* Nov 2022
Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Pulkit Agrawal's lab

Learning Mobile Manipulation, *Boston Dynamics* Nov 2022
Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects

Learning Mobile Manipulation, *Boston Dynamics AI Institute* Nov 2022
Presented a collection of works done during my Ph.D.

Learning Mobile Manipulation, *MITRE* Nov 2022
Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects

Learning Mobile Manipulation, *Rotor.ai* Oct 2022
Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects
Learning Mobile Manipulation, *STR* Oct 2022
Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects
Mobile Manipulation Leveraging Multiple Views, *IROS 2022* Oct 2022
Presented work on localization free mobile manipulation that was nominated for best paper in mobile manipulation
P.h.D. Defense: Learning Mobile Manipulation, *Columbia University* May 2022
Presented my dissertation defense enabling mobile robots to manipulate objects
Minecraft Basalt Interview, *Yannic Kilcher YT* Jan 2022
Interviewed on recent first place win in Minecraft Basalt RL competition.
Learning Mobile Manipulation, *ROAM Lab* Dec 2021
Presented work done in preparation of defense of my thesis to the ROAM lab at Columbia University.
Learning Mobile Manipulation, *CAIR Lab* Feb 2021
Presented my thesis proposal to the CAIR lab at Columbia University.
IROS 2020: Learning Your Way Without Map or Compass, *IROS 2020* Oct 2020
Presented my Learning Your Way work at IROS 2020 via the online conference.
Demystifying the Dissertation, *Columbia University* Jun 2020
Presented my thesis proposal and help current graduate students understand the process of completing a PhD.
Thesis Proposal: Learning Mobile Manipulation, *New York, NY* Jan 2020
Presented and defended my thesis proposal in learning mobile manipulation in January 2020.
Research Talk, *Harlem Children's Zone STEM Exposure* Nov 2019
Shared my research as part of an initiative to help teach children in Harlem methods in STEM research.
Learning Your Way Talk, *NYU Reading Group* Oct 2019
Presented Learning Your Way work to the NYU robotics reading group.
Learning Your Way Talk, *NERC 2019* Oct 2019
Presented Learning Your Way work in front of audience of robotics researchers at NERC 2019 conference.
Visual Tactile Completion, *Emptor Lightning Talks* Sep 2019
Presented work on visual tactile grasping as well as next steps as part of Emptor's lightning talks.
Odefi Pitch, *Columbia IBM Blockchain Accelerator Demo Day 2019* May 2019
Enabling credit default swaps on the ethereum network pitched to investors at the capstone event for the Columbia IBM Blockchain Accelerator.
Visual Tactile Grasping, *Samsung Research NYC* July 2019
Presented work on visual tactile grasping as well as next steps.
Candidacy Exam: Simulation for Real World Robotics, *New York, NY* May 2019
A high-level overview of how real-world robotics can be enabled through simulation.
Using Simulation to Enable Generated Art and Robotics, *Making Art in the Age of Algorithms* Dec 2018
A high-level overview of how robotics can be enabled through simulation as part of a series of lightning talks about art and algorithms.
Visual Tactile Completion Poster, *Data Science Day 2018* Mar 2018
This work provides an architecture that incorporates depth and tactile information to create rich and accurate 3D models useful for robotic manipulation tasks presented at Data Science Day 2018 at Columbia University.
Lecture: ROS Tutorial, *New York, NY* Jan 2018
An introductory tutorial on ROS and use of robotics in the Columbia Robotics Lab to aspiring roboticists.
Providing Context to Startup Culture, *New York, NY* May 2016
An analysis on the effectiveness of a startup based on the type of culture it maintains as well as effects on profit/loss.

Memberships

ACM 2016 — Present
IEEE 2016 — Present
SHPE 2018 — Present
AAAI 2022 — Present

Teaching Experience

Humanoid Robotics (COMSW 6731), Teaching Assistant, Graduate Level Spring 2018
Computational Aspects of Robotics (COMSW 4733), Teaching Assistant, Graduate Level Fall 2017
Programming Languages and Translators (COMSW 4115), Teaching Assistant, Graduate Level Fall 2016
Object Oriented Programming and Design in Java (COMS 1007), Teaching Assistant, Undergraduate Level Fall 2016
Programming Languages and Translators (COMSW 4115), Teaching Assistant, Graduate Level Spring 2016
Object Oriented Programming and Design in Java (COMS1007), Teaching Assistant, Undergraduate Level Fall 2014
Fundamentals of Computer Systems (CSEE3827), Teaching Assistant, Undergraduate Level Spring 2014
Object Oriented Programming and Design in Java (COMS1007), Teaching Assistant, Undergraduate Level Fall 2013

References

References available upon request.